Phylogenetic Trees

BioInf 2025/2026 1st Semester

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Recap

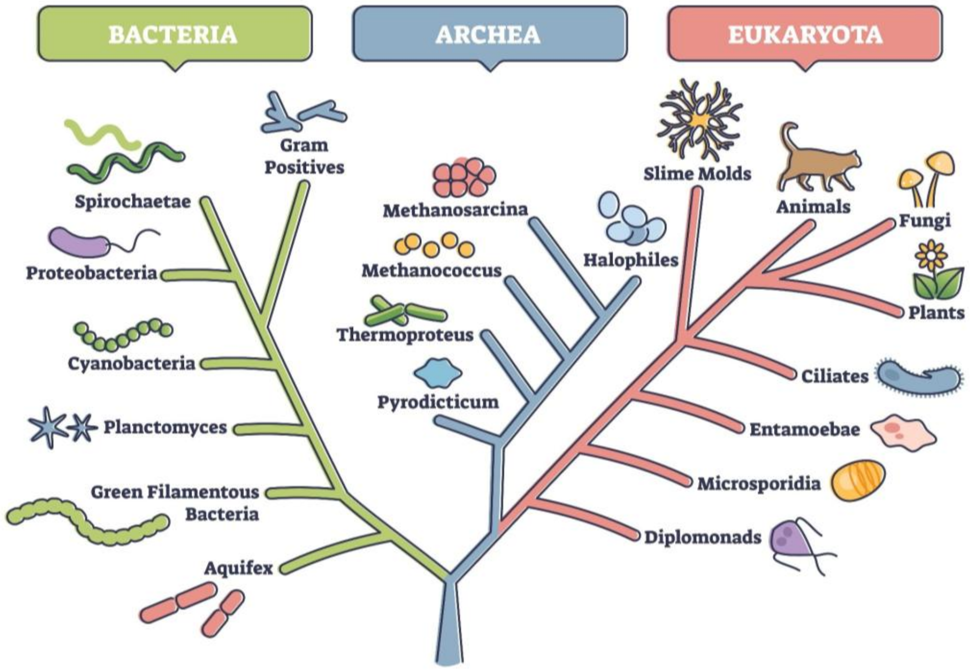
- Biological data is sequential (DNA, RNA, proteins)
- Biological processes transform DNA->RNA->proteins
- Algorithms to process sequences and simulate biological processes
- Algorithms to find patterns in biological sequences
- Algorithms to align sequences to maximize an alignment score

Recap

- Global Alignment: Maximize overall matches
- Local Alignment: Maximize concentrated matches
- Semi-global: Disregard start and/or end gaps
- Database search: find matches with heuristics
- Multiple Sequence Alignments: find what multiple sequences have in common

Phylogenetic Trees

- **Phylogenetics**: branch of Biology which studies the **evolutionary** history and relationships among species.
- Determination of how a given set of sequences has **evolved from a common ancestor** through a process of natural evolution.
- Evolutionary tree: bifurcations represent events of **mutation** from a common ancestor, that give rise to two (or more) different branches.



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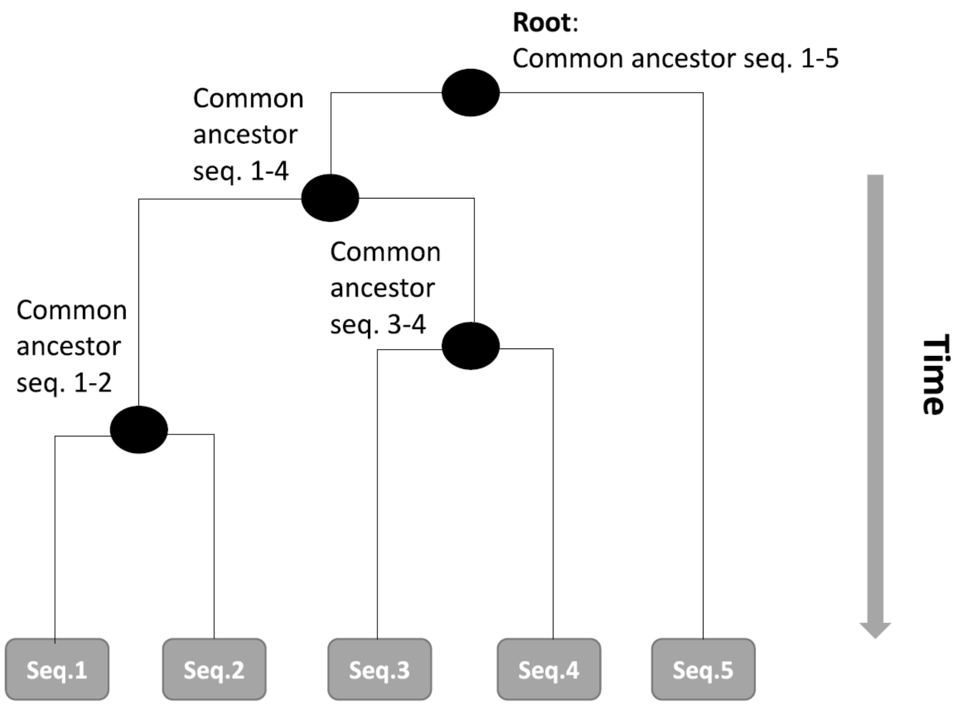
Phylogenetic Trees

- Leaf nodes: known sequences (DNA, RNA, protein)
- Internal nodes: common ancestors of the sequences below.
- Root of a tree: common ancestral of all sequences (taxa).



- Height of the nodes in the tree: measure of time.
- Traveling from the root to the leaves represents the passage of time.

Phylogenetic Trees

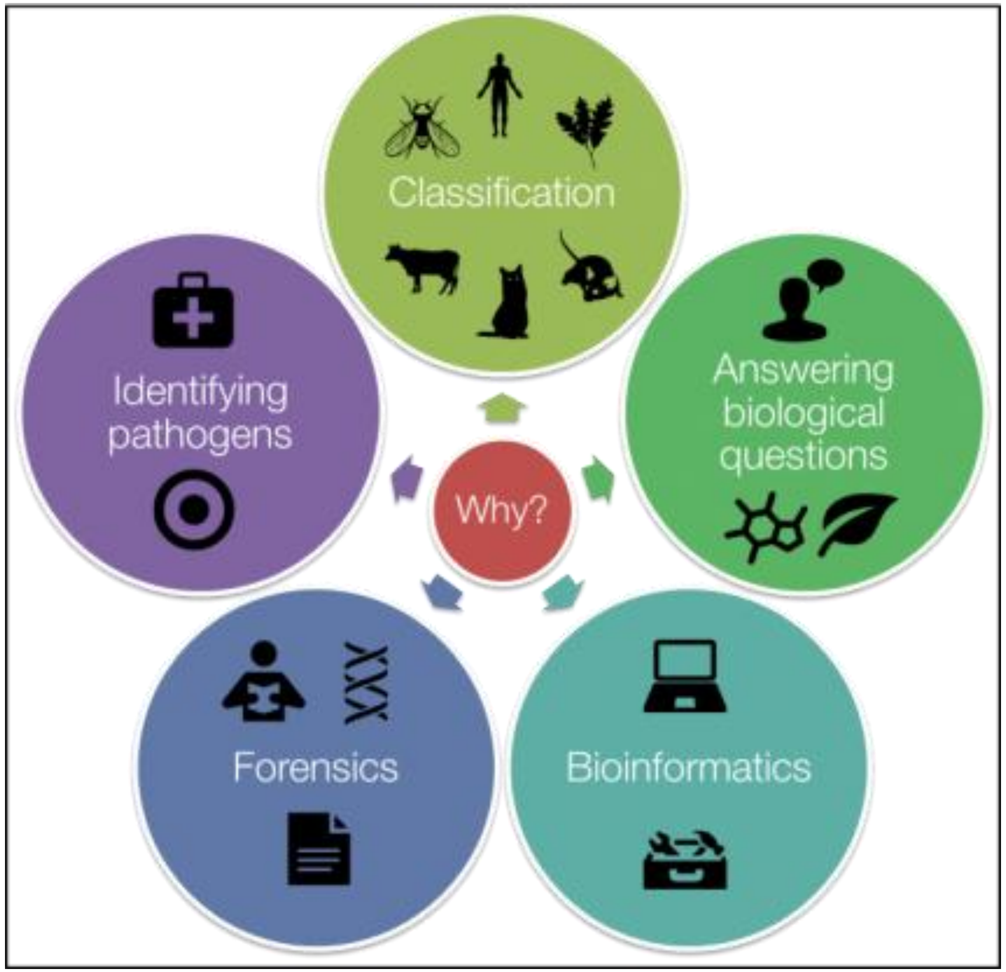


Image source: EMBL-EBI

Phylogenetic Trees

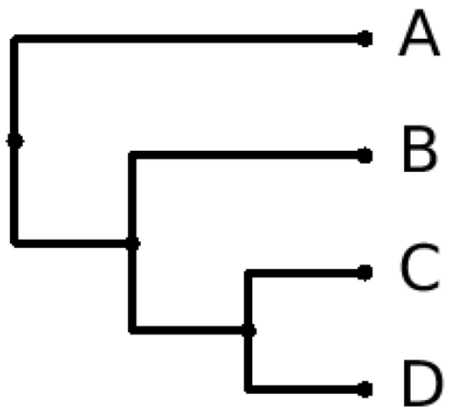
What is a tree?

- Connected acyclic undirected graph
  - Graph: mathematical structure composed of nodes (or vertices or points) connected by edges (or arcs or lines)
  - Connected: all nodes are connected by some path
  - Acyclic: no loops
  - Undirected: the edges do not have a direction
- Any two nodes are connected by a unique path
  - A leaf is a node connected to only one other node

Newick Format

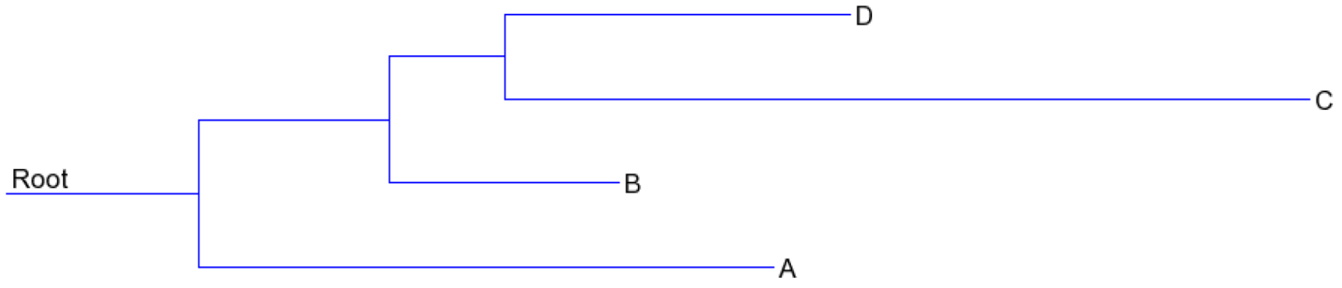
- Newick is a standard format for phylogenetic trees
    - (branch_1, … , branch_n) ;
    - (means new branch
    - ; marks end of tree
    - each branch is one tree

- (A, (B, (C,D)));



Newick Format

- Newick is a standard format for phylogenetic trees
  - (branch_1, ... , branch_n) ;
  - ; marks end of tree
  - each branch is one tree
- We can include branch lengths
  - (A:5, (B:2, (C:7, D:3):1):2);



Phylogenetic Trees

The problem of building a phylogenetic tree

- Organize a **set of taxa** (species, genes, proteins or such elements) in a way that:
  - **each leaf** corresponds to **each taxon**
  - the other nodes of the trees represent **ancestral species**
- Tree should **reflect the evolutionary relations** between the taxa
- Optimization problem: for a given set of sequences there are several alternative trees that could explain their evolution from a common ancestor
  - the number of possible trees grows very rapidly, even for a moderate number of sequences in the input
  - for 30 sequences there are around 10^{40} different trees!

Methods for Computing Phylogenetic Trees

- **Distance-based**:
  - based on a **matrix of pairwise distances** among the sequences, seeking to find the trees where the **distances are consistent** with the ones in the input matrix;
- **Maximum parsimony**:
  - search for the trees where the **number of needed mutations** (in internal nodes of the tree) to explain the variability of the sequences is **minimized**.
- **Maximum likelihood**:
  - probabilistic models for the occurrence of different types of mutations, and uses those models to **score trees based on their probability**, searching for the **most likely trees** that explain the sequences according to the assumed model.

Methods for Computing Phylogenetic Trees

Distance-based Methods

- Objective functions that measure the consistency of the **distances between the leaves** (representing sequences) in the tree, with the ones **obtained from sequence similarity** (through sequence alignment).
- Try to adjust the **structure** of the tree and the **length** of the edges connecting the nodes to mimic the pairwise distances between sequences.
- Given a set of sequences and a distance matrix, build a tree where the arcs between two taxa correspond to the distances.

Methods for Computing Phylogenetic Trees

Distance-based Methods

- The pairwise alignments used before obtaining the MSA are not adequate for reliable phylogenetic trees.
- Typically, when building phylogenetic trees from sequence data we compute the MSA first and then measure distances using the MSA.
- Simple approach; calculate the percentage of columns where the MSA contains mismatches or gaps.

Methods for Computing Phylogenetic Trees

Distance-based Methods

- Example: alignment of five primate nucleotide

```
Human       GTAAATATAGTTTAACCAAAACATCAGATTGTGAA    35
Chimpanzee  .....t.........c........c..........    35
Gorilla     .....t.........c........c..........    35
Orangutan   .....t.........c........t..........    35
Gibbon      .....c.........t........t..........    35

Human       TCTGACAACAGAGGCTTACGACCCCTTATTTACCG    70
Chimpanzee  ...g.c..c..a.g.tcacg...c...at......    70
Gorilla     ...g.t..c..a.g.tcaca...c...at......    70
Orangutan   ...a.t..t..g.c.ccaca...c...at......    70
Gibbon      ...a.c..t..a.g.tcgaa...t...gc......    70
```

- Distances:

1/70

3/70

	Human	Chimpanzee	Gorilla	Orangutan	Gibbon
Human	-				
Chimpanzee	0.014	-			
Gorilla	0.043	0.029	-		
Orangutan	0.129	0.114	0.086	-	
Gibbon	0.171	0.157	0.157	0.157	-

Methods for Computing Phylogenetic Trees

Distance-based Methods

- Given the matrix of distances, minimize differences between distances in the tree and in the matrix:

$$score(T) = \sum_{i,j \in S} \left(d_{i,j}(T) - D_{i,j} \right)^2$$

Methods for Computing Phylogenetic Trees

Distance-based Methods

- The distance $d_{i,j}$ between nodes i and j in a phylogenetic tree T is the **vertical distances traveled from the origin *i* to the destination *j***
- If k is the nearest common ancestor of i and j, then:

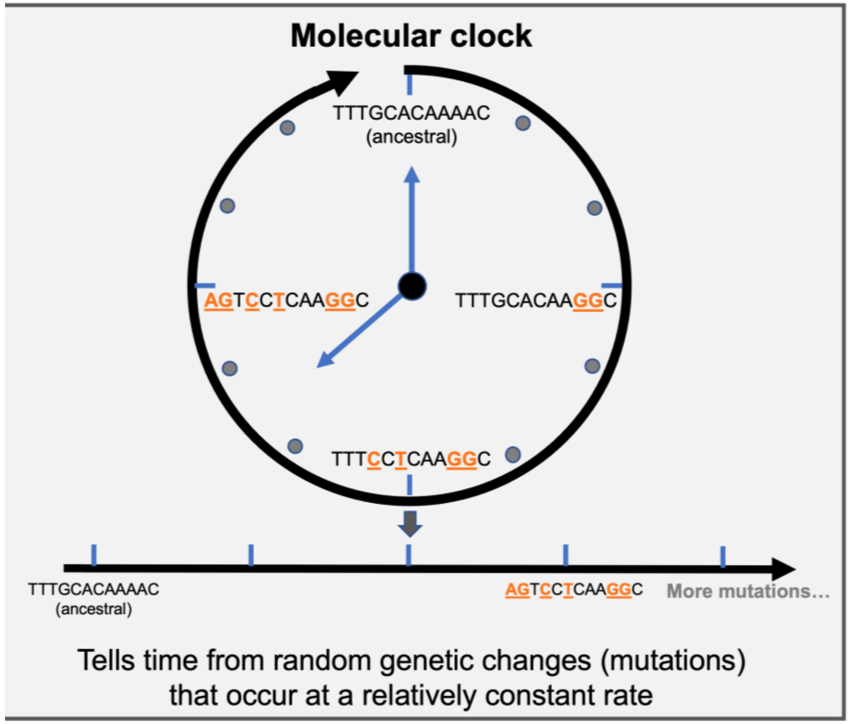
$$d_{i,j} = d_{i,k} + d_{k,j}$$

- Both these distances are computed by the difference of the heights of the nodes:

$$d_{i,k} = h(k) - h(i) \text{ and } d_{k,j} = h(k) - h(j)$$

- Assuming uniform mutation rate in all branches of the tree: distance between all leaves and the root is the same and the height of each leaf can be defined as 0:

$$d_{i,j} = 2h(k)$$



https://www.czbiohub.org/rapid-response/genomic-epidemiology/sequence-alignments-molecular-clocks/

UPGMA

UPGMA: Unweighted Pair Group Method with Arithmetic Averages

1. Consider each sequence (tree leaf) to be in **its own cluster**, which are put at **height zero** in the tree.
2. Join the pair of **closest sequences/clusters** (i.e. the minimum value in the matrix D), creating an internal node, **with height in the tree equal to half of the distance between these sequences**.
3. In the next iteration, consider those **sequences as a single cluster**
    - remove columns and rows of the connected sequences
    - add new row and new column for the new cluster
    - the distances to the remaining calculated by the average of previous the distances
4. Repeat until when all sequences are **within a single cluster**, which corresponds to the root node of the tree.

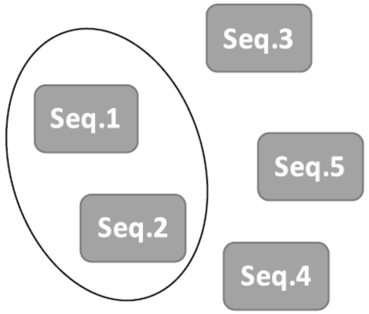
UPGMA

- Example:

Distance matrix:

	S1	S2	S3	S4
S2	2			
S3	5	4		
S4	7	6	4	
S5	9	7	6	3

	S1, S2	S3	S4
S3	4.5		
S4	6.5	4	
S5	8	6	3



Clusters

Tree

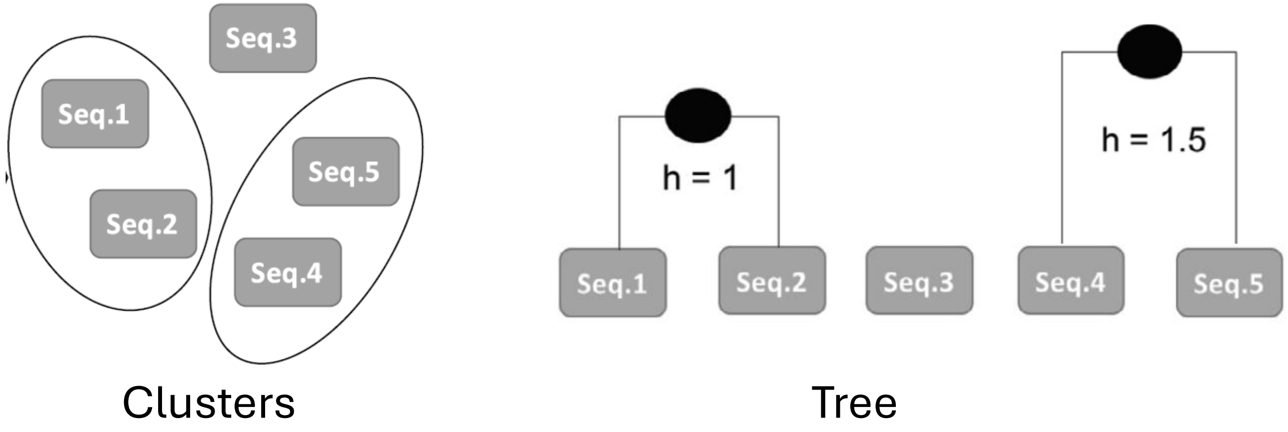
UPGMA

- Example:

Distance matrix:

	S1, S2	S3	S4
S3	4.5		
S4	6.5	4	
S5	8	6	3

D	S1, S2	S3
S3	4.5	
S4, S5	7.25	5



UPGMA

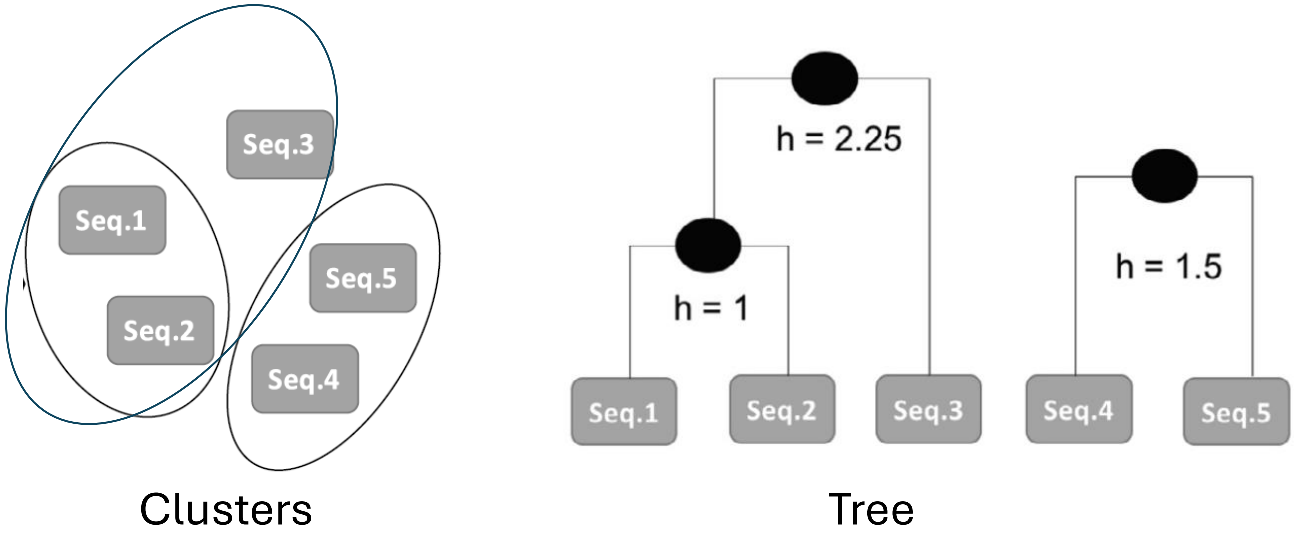
- Example:

Distance matrix:

D	S1, S2	S3
S3	4.5	
S4, S5	7.25	5

D	S1, S2,S3
S4, S5	6.5

$$\frac{7.25 + 5}{2} = 6.125$$

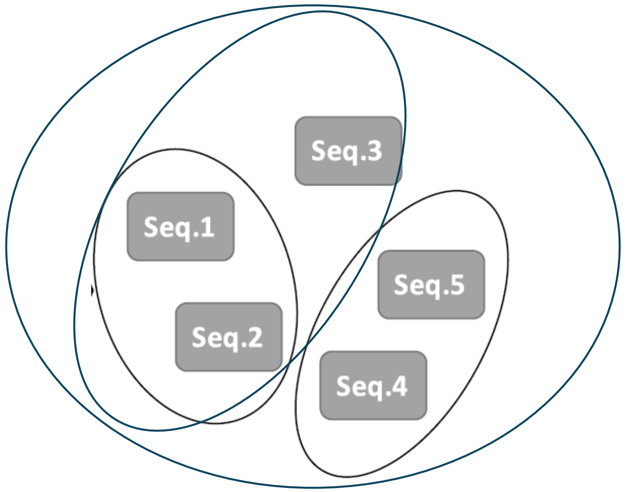


UPGMA

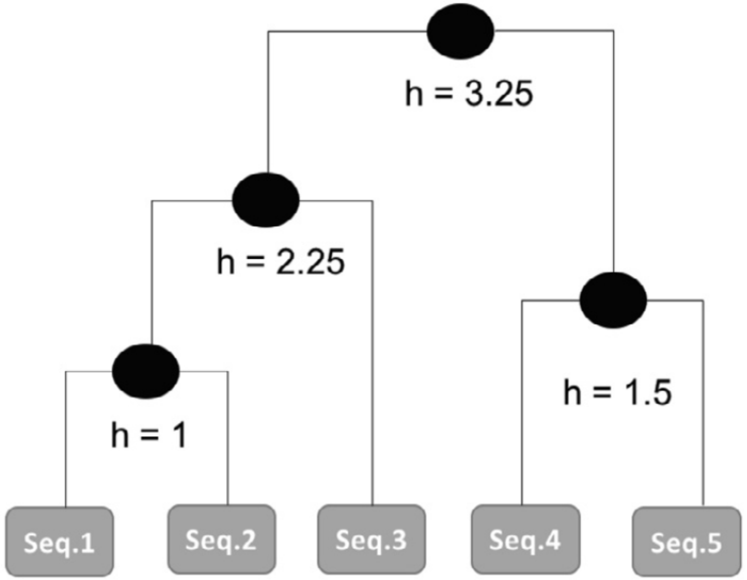
- Example:

Distance matrix:

D	S1, S2,S3
S4, S5	6.5



Clusters



Tree

UPGMA

- Distance between two clusters A and B:

$$D(A,B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} D_{ij}$$

- Distances from the new cluster ($A \cup B$), to each cluster X:

$$D(A \cup B, X) = \frac{|A|D(A,X) + |B|D(B,X)}{|A| + |B|}$$

UPGMA

- Example:

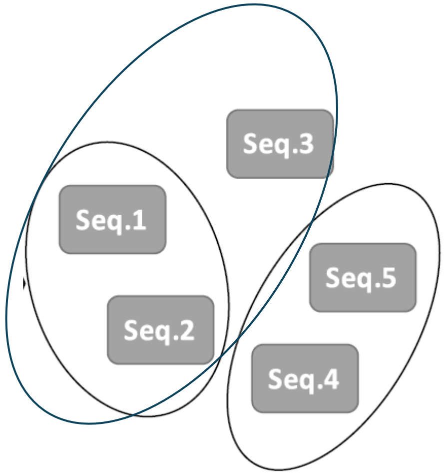
Distance matrix:

D	S1, S2	S3
S3	4.5	
S4, S5	7.25	5

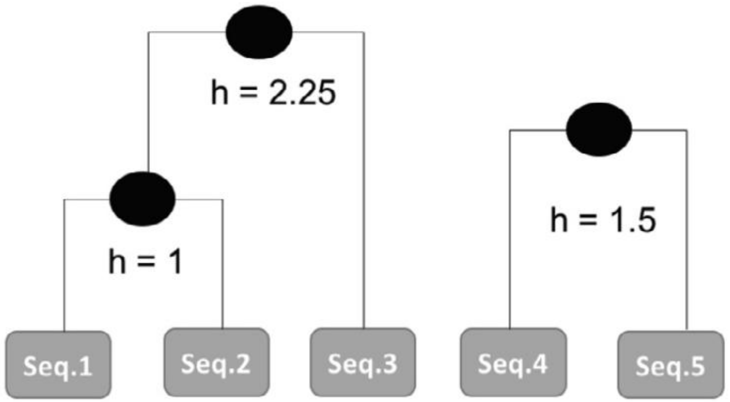
D	S1, S2,S3
S4, S5	6.5

$$\frac{7.25 + 5}{2} = 6.125$$

$$\frac{2 * 7.25 + 1 * 5}{2 + 1} = 6.5$$



Clusters



Tree

Distance-based Methods

UPGMA: Unweighted Pair Group Method with Arithmetic Averages

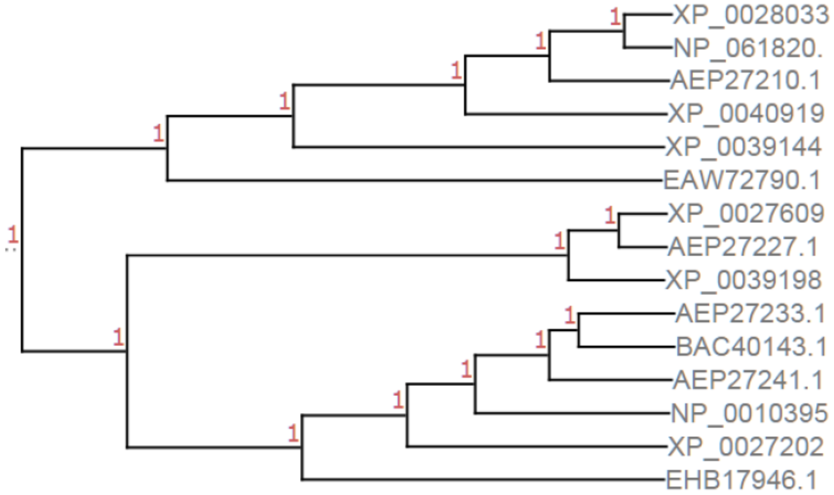
- The tree branches all have same total length
- Assumes constant mutation rate

Neighbour-Joining

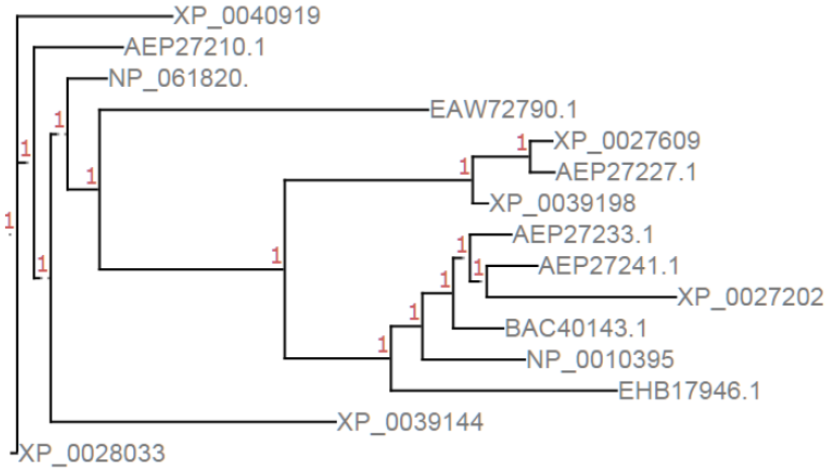
- Minimizes total branch length but branch lengths may differ
- Assumes minimal evolution

Distance-based Methods

UPGMA



Neighbour-Joining



Neighbour-Joining

- Criterion to select the clusters to merge in each step does not only take into account the distances between the clusters, but also seeks to select **cluster pairs where the nodes are apart from the others**
- Original distance matrix D is pre-processed to create a new matrix Q, where each cell is calculated as:

$$Q_{ij} = (n-2)D_{ij} - \sum_{k=1}^{n} D_{ik} - \sum_{k=1}^{n} D_{jk}$$

- It will be this matrix Q that will be used to select the nearest clusters in each step, as it is the case in UPGMA

Neighbour-Joining

- Taking u as the new cluster (new node in the tree) created joining nodes a and b, the distances from the other clusters/nodes (represented as i) are calculated as:

$$D_{ui} = \frac{1}{2}(D_{ai} + D_{bi} - D_{ab})$$

- From this new matrix D, the new matrix Q is recalculated applying the expression given above, and the algorithm proceeds by choosing the minimum value to select the clusters to merge

Maximum Parsimony

- Objective function: estimate the number of mutations that are implied by the tree to explain the input sequences:
  - shorter trees that explain the data are preferred (simpler models)
- Can be based on the MSA of the input sequences, which allows to identify columns where there are mutations (gaps or mismatches) and thus variations of the different sequences.
  - From this, the method tries to identify the tree that requires the minimum number of mutations to explain the variations.
  - Does not scale well (up to 10 sequences)
- Can also use branch and bound methods or heuristic methods for more efficiency (20 sequences or more)

Maximum Likelihood

- Estimate the likelihood of possible trees, by multiplying the estimated probability of each mutation implied by the tree.
  - Requires a statistical model of the occurrence of mutations in the DNA
- Great flexibility given the plethora of possible mutation models that can be considered.
- Typically quite intensive computationally.
- The pruning algorithm, a variant of Dynamic Programming can be used by computing the likelihood of sub-trees, thus decomposing the overall problem in smaller sub-problems.

Summary

- Phylogenetic Trees
- Methods for Computing Phylogenetic Trees
  - Distance-based Methods: UPGMA, Neighbour-Joining
  - Maximum Parsimony
  - Maximum Likelihood

Further reading

- Rocha Chapter 8